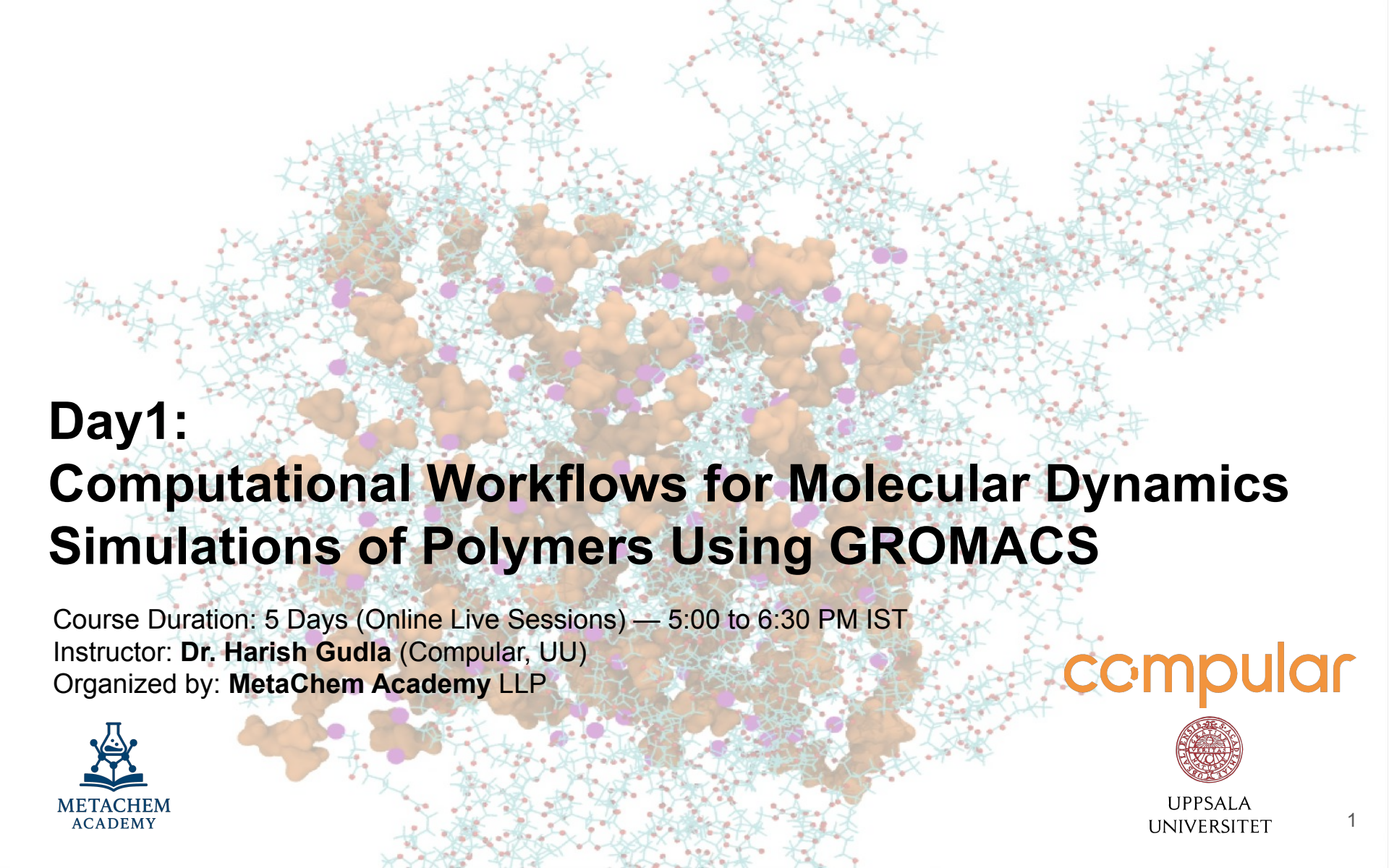




Day1: Computational Workflows for Molecular Dynamics Simulations of Polymers Using GROMACS

Course Duration: 5 Days (Online Live Sessions) — 5:00 to 6:30 PM IST

Instructor: **Dr. Harish Gudla** (Compular, UU)

Organized by: **MetaChem Academy LLP**



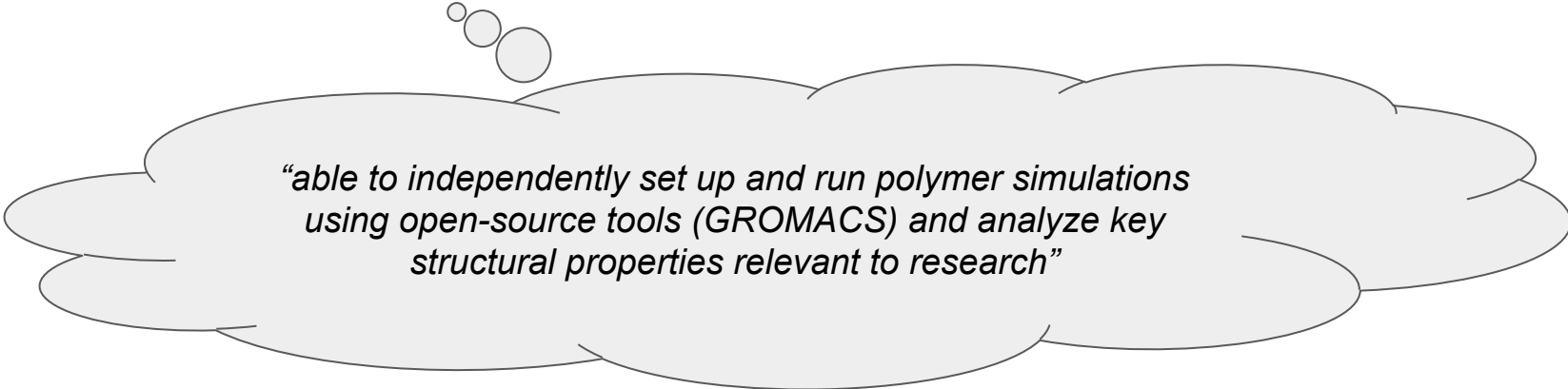
METACHEM
ACADEMY

compular



UPPSALA
UNIVERSITET

About the Workshop



“able to independently set up and run polymer simulations using open-source tools (GROMACS) and analyze key structural properties relevant to research”

Dates
10 - 16 Dec

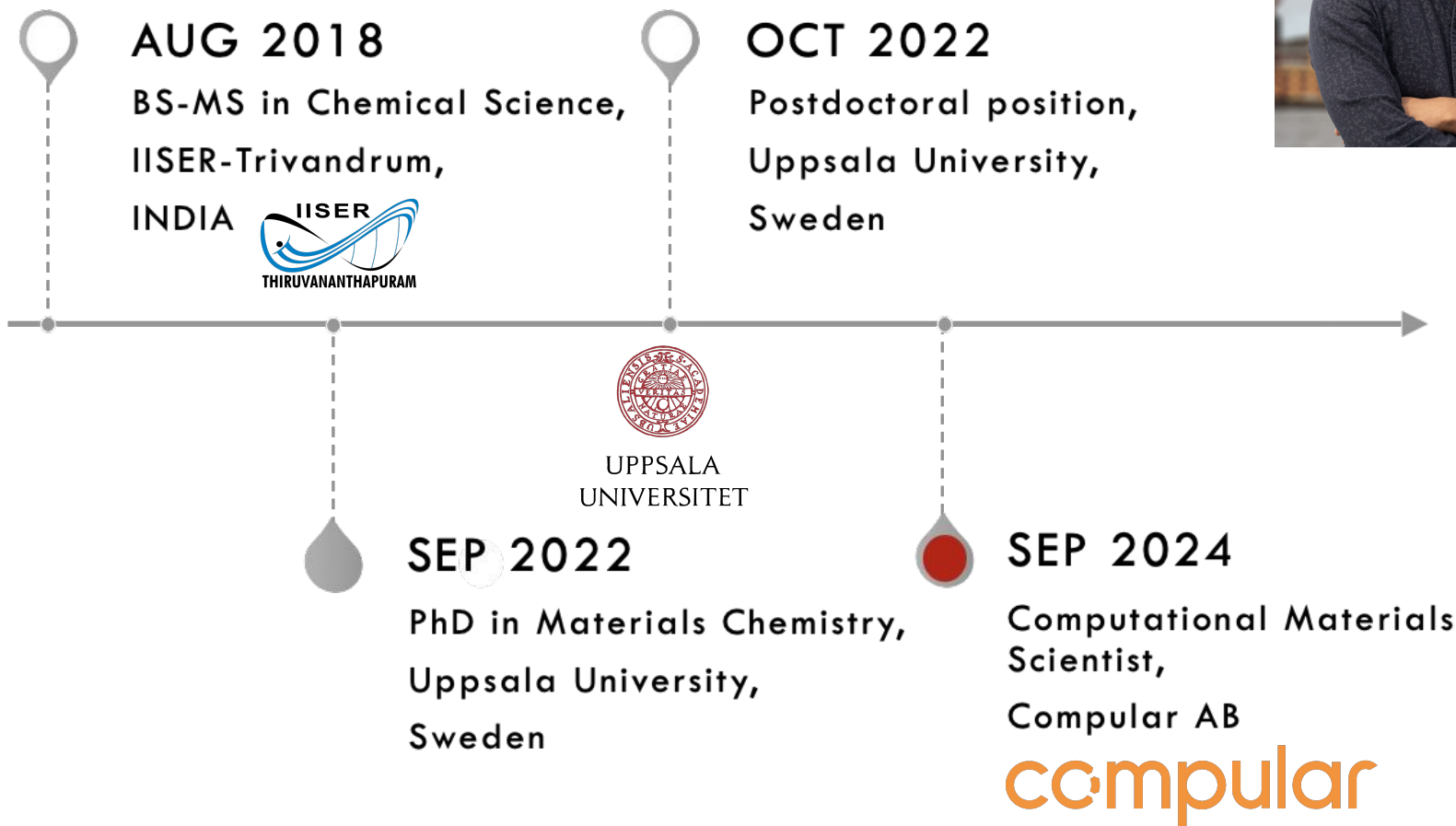
Theory Session (5:00–5:30 PM IST)

Hands-On Session (5:30–6:30 PM IST)

Few things to keep in mind:

- Feel free to interrupt anytime with questions.
- Session's pace can be adjusted according to whole group.
- Learning opportunities

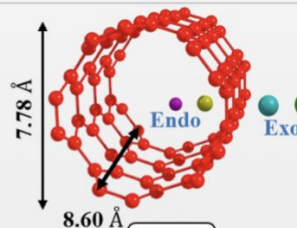
Who am I?



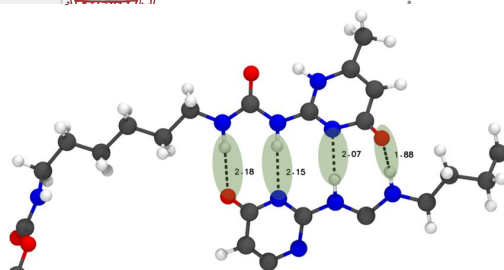
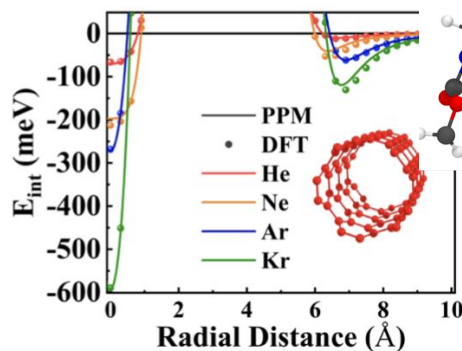
Who am I?

AUG 2018

BS-MS in Chemical Science,
IISER-Trivandrum,
INDIA



Fit 2



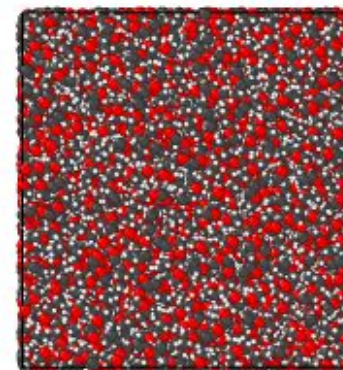
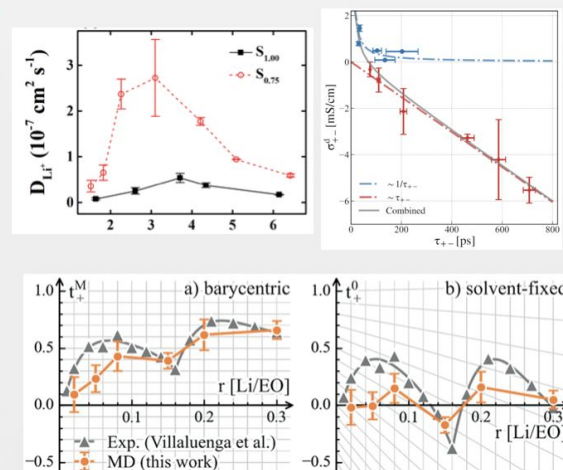
versity,

C

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GROMACS, LAMMPS

Python codes for simulation workflows and analysis



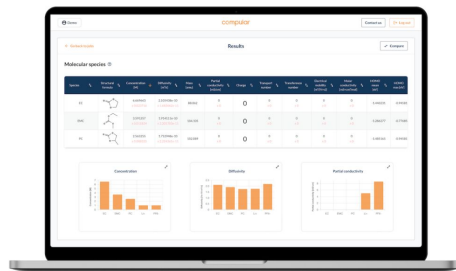
What I do?

Computational Material Scientist

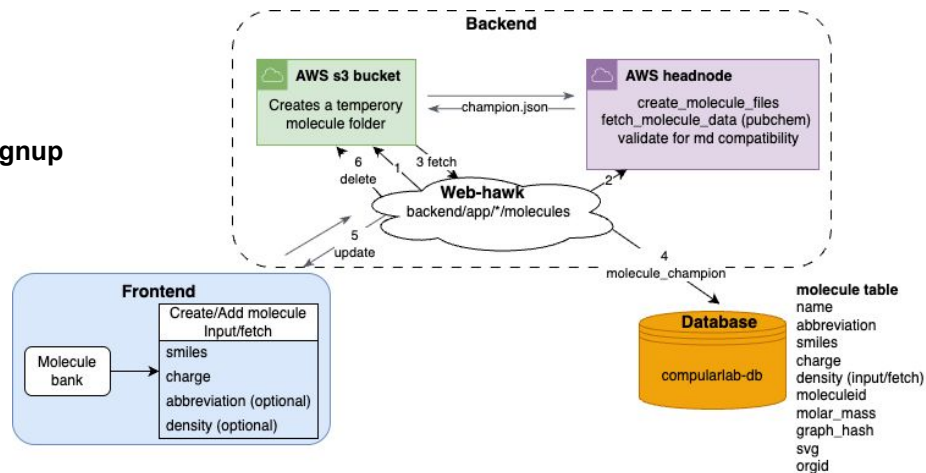
- Developing automated, modular simulation workflows for new features in Compular's SaaS platform (**Compular Lab**)
- Running multiple industrial R&D projects
- Deploying and managing cloud-based HPCs (AWS, Azure) to perform large-scale simulations and analyze complex material properties.
- Exploring new application areas, including polymer membranes for fuel cells,
- Designed and scripted automation steps and workflow orchestration concepts to streamline simulation pipelines (Python, Bash).

Welcome to Compular

Digitalising material development



Try Compular Lab's Demo:
<https://compularlab.com/signup>



Who are you?



What is your profession

Multiple Choice Poll

✓ 17 votes

👤 17 participants

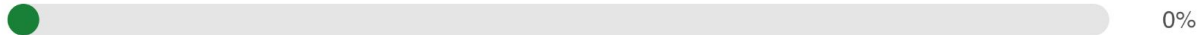
Bachelors/Masters - 5 votes



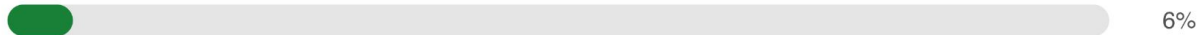
PhD - 11 votes



Postdoc/Researchers - 0 votes



Others - 1 vote



Who are you?



Which best describes your current research toolbox when studying polymers?

Multiple Choice Poll



18 votes



18 participants

Computational tools (DFT, MD, etc.) - 6 votes



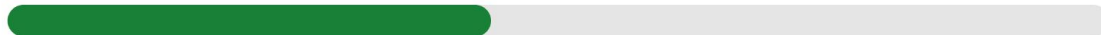
33%

Experimental — synthesis, characterization, measurements - 7 votes



39%

Mix of Both — use experiments and simulations together - 8 votes



44%

Not much experience with polymers - 3 votes



17%

Who are you?



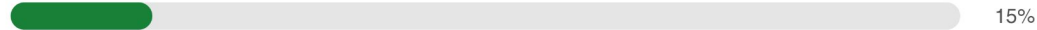
Familiarity with tools/software we cover in this workshop

Multiple Choice Poll

✓ 20 votes

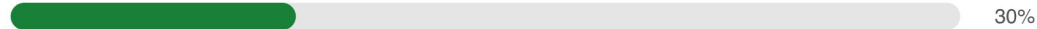
👤 20 participants

Github - 3 votes



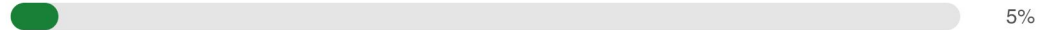
15%

Python/Jupyter notebooks - 6 votes



30%

Ambertools (or others) - 1 vote



5%

GROMACS (or LAMMPS) - 7 votes



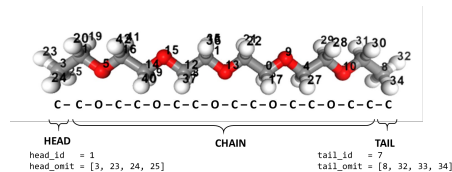
35%

None - 10 votes

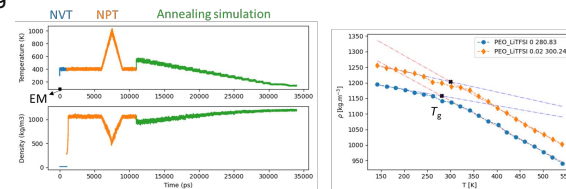
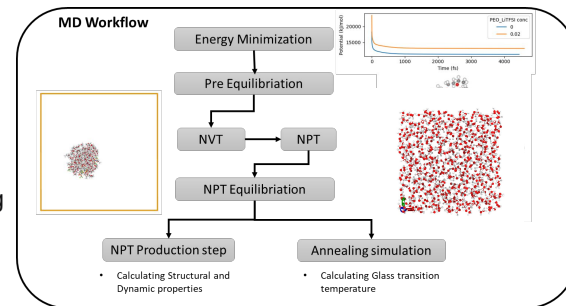
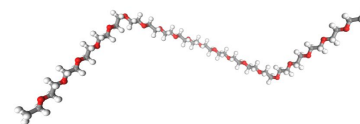


50%

Workshop Schedule



Date	Theory Session (5:00–5:30 PM IST)	Hands-On Session (5:30–6:30 PM IST)
10 Dec	Introduction to classical MD, force fields, and motivation for polymer simulations	Installing and setting up tools and Parameterizing monomers
11 Dec	Polymer chain architecture: HEAD/CHAIN/TAIL atom definitions and polymerization logic	Building polymer chains using parameterized monomers and preparing input files
12 Dec	System preparation: multi-chain packing, topology files, and GROMACS input formats	Generating multi-chain systems, topology, and coordinate files ready for MD
15 Dec	MD equilibration principles: Energy Minimization, NVT, NPT ensembles, and monitoring stability	Running equilibration simulations and analyzing density and RMSD for structural stability
16 Dec	Thermal and conformational analysis: annealing simulations and glass transition temperature (Tg) concepts	Performing annealing runs and calculating polymer properties: density, RMSD, Rg, and Tg



Day 1 - Outline

- Theory session (5:00-5:45 pm IST)
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 - Force Field parameters
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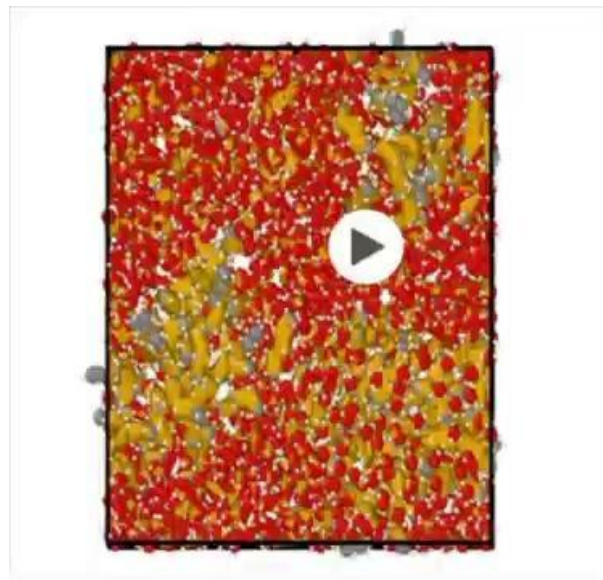
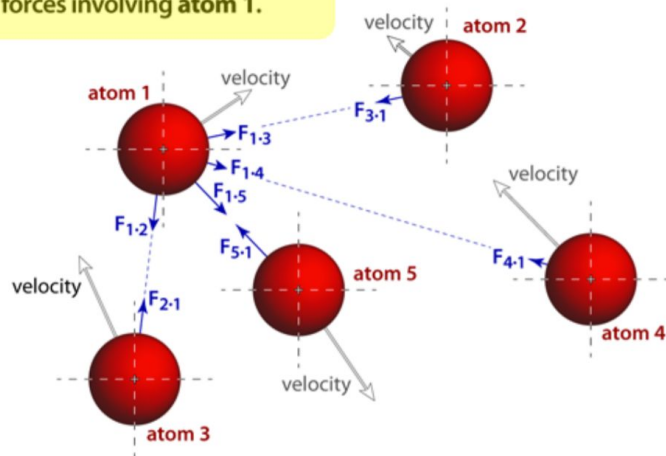
Introduction to Molecular Dynamics

“everything that living things do can be understood in terms of the **jiggling and wiggling** of atoms.”
-The Feynman Lectures in Physics vol. 1, 3-6 (1963)

Numerically integrating Newton's equations of motion

$$m_i \frac{\partial^2 \mathbf{r}_i}{\partial t^2} = \mathbf{F}_i = -\partial V / \partial \mathbf{r}_i$$

Calculate the interatomic forces involving atom 1.



Water
+
Oil

MD algorithm

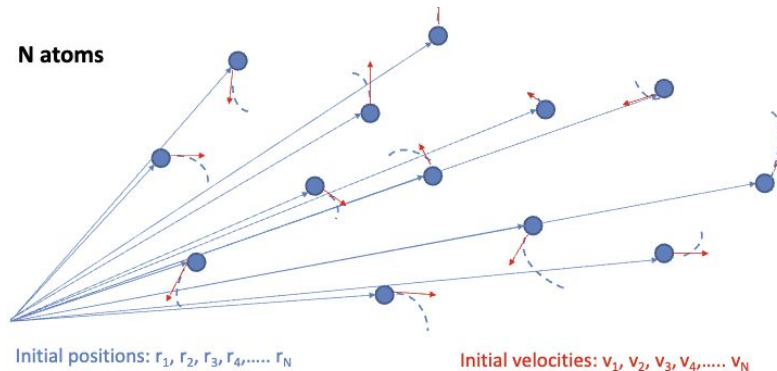
Step 1: initial positions $r^{(i=0)}$ and velocities $v^{(i=0)}$
for atoms, set $a = 0$, $t = 0$, $i = 0$, timestep Δt

Step 2: after atom movements, location update: $r^p = r^i + V^{(i)}\Delta t + \frac{1}{2}a\Delta t^2 + \text{accurate terms}$, velocity update:
 $v^p = V^{(i)} + a\Delta t + \text{accurate terms}$

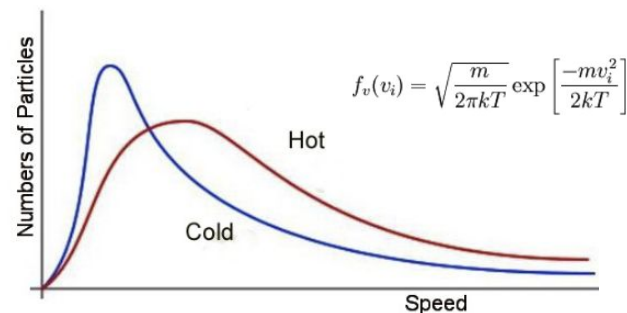
Step 3: force/energy calculations: $F = -\nabla v(r^p)$ or $F = F(\Psi(r^p))$ and $a = F / m$

Step 4: after atom movements, location update: $r^{i+1} = r^p + f(a, \Delta t)$, velocity update: $v^{(i+1)} = v^p + f(a, \Delta t)$

Step 5: Apply boundary conditions, temperature and pressure control. Time integration and iteration step: $t = t + \Delta t$, $i = i + 1$. Repeat step 2 until the acceptable results are obtained.



Boltzmann distribution of initial velocities



MD algorithm

Step 1: initial positions $r^{(i=0)}$ and velocities $v^{(i=0)}$
for atoms, set $a = 0$, $t = 0$, $i = 0$, timestep Δt

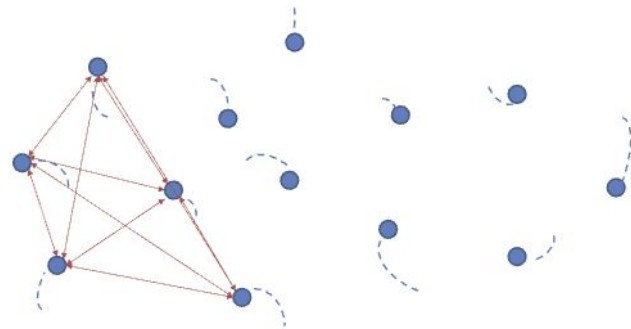
Step 2: after atom movements, location update: $r^p = r^i + V^{(i)}\Delta t + \frac{1}{2}a\Delta t^2 + \text{accurate terms}$, velocity update:
 $v^p = V^{(i)} + a\Delta t + \text{accurate terms}$

Step 3: force/energy calculations: $F = -\nabla v(r^p)$ or $F = F(\Psi(r^p))$ and $a = F / m$

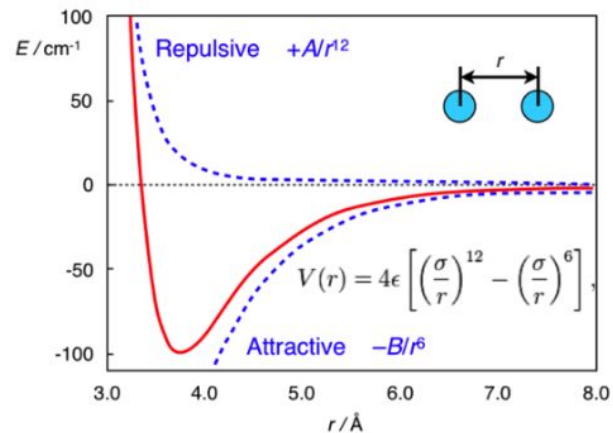
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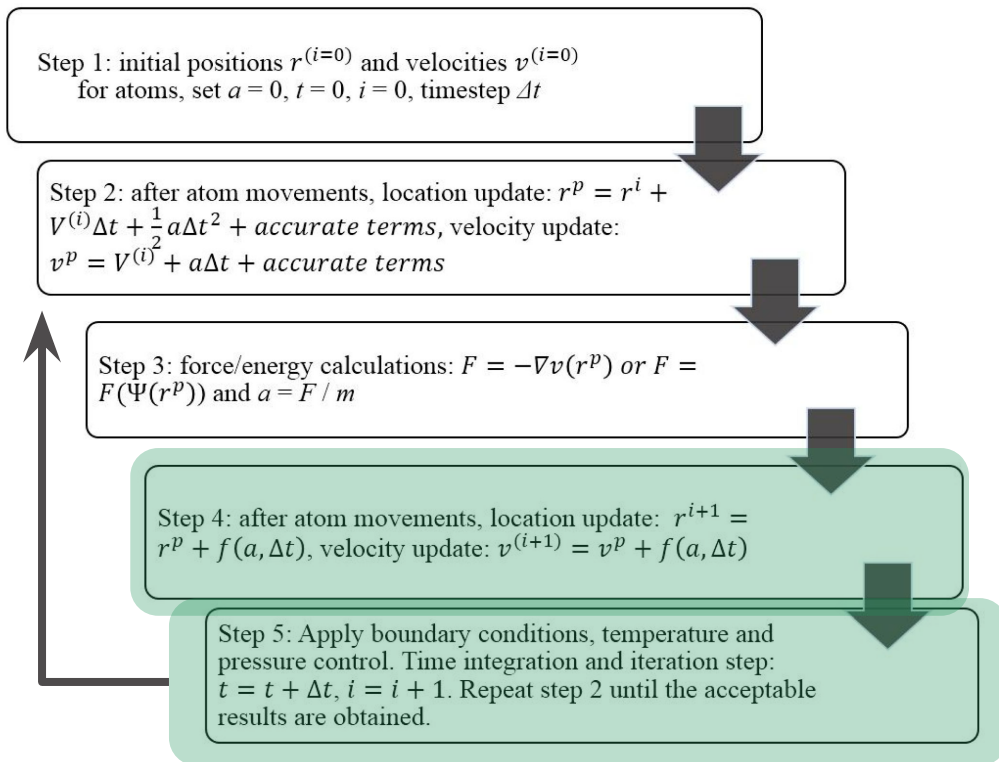
N atoms



Interatomic pair potentials: $U_{1,2}, U_{1,3}, U_{1,4}, U_{1,5}, U_{2,3}, U_{2,4}, U_{2,5}, \dots, U_{N,N-1}$



MD algorithm



Verlet algorithm

- update positions and velocities

$$\mathbf{r}(t + \delta t) = \mathbf{r}(t) + \mathbf{v}(t)\delta t + \frac{1}{2}\mathbf{a}(t)\delta t^2 + \dots$$
$$\mathbf{r}(t - \delta t) = \mathbf{r}(t) - \mathbf{v}(t)\delta t + \frac{1}{2}\mathbf{a}(t)\delta t^2 + \dots$$

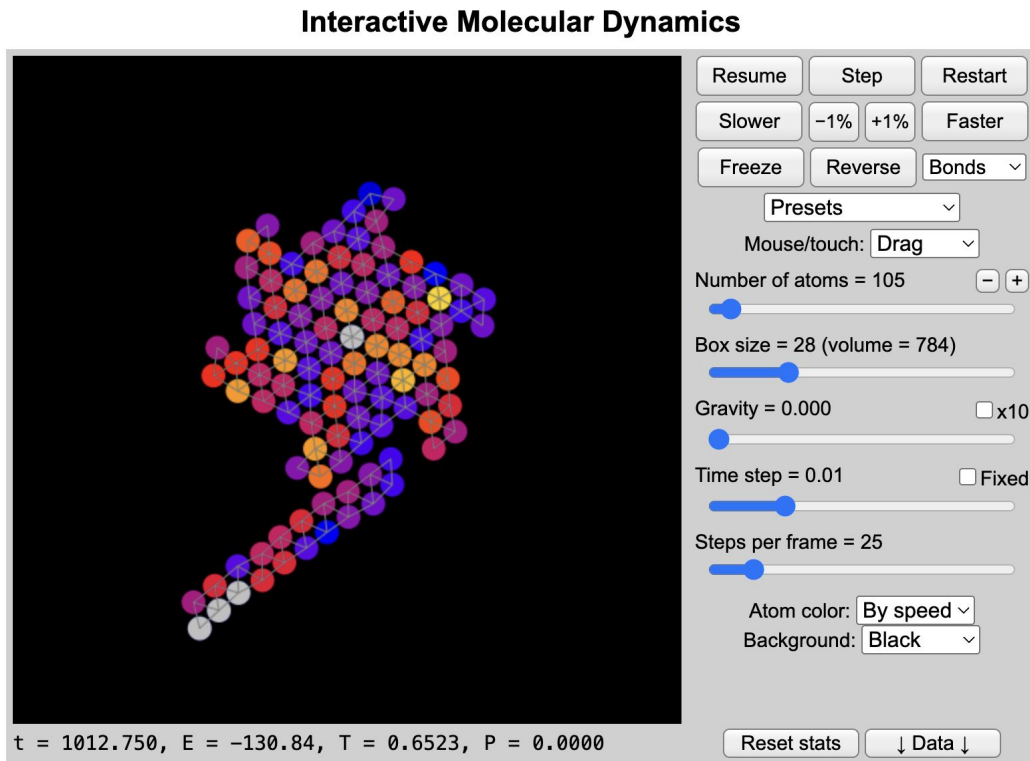
$$\mathbf{r}(t + \delta t) = 2\mathbf{r}(t) - \mathbf{r}(t - \delta t) + \mathbf{a}(t)\delta t^2$$

$$\mathbf{v}(t + \delta t) = \frac{\mathbf{r}(t + \delta t) - \mathbf{r}(t - \delta t)}{2\delta t}$$

Simple interactive MD

Web app simulates the dynamics of simple atoms and molecules in a two-dimensional universe

<https://physics.weber.edu/schroeder/md/>



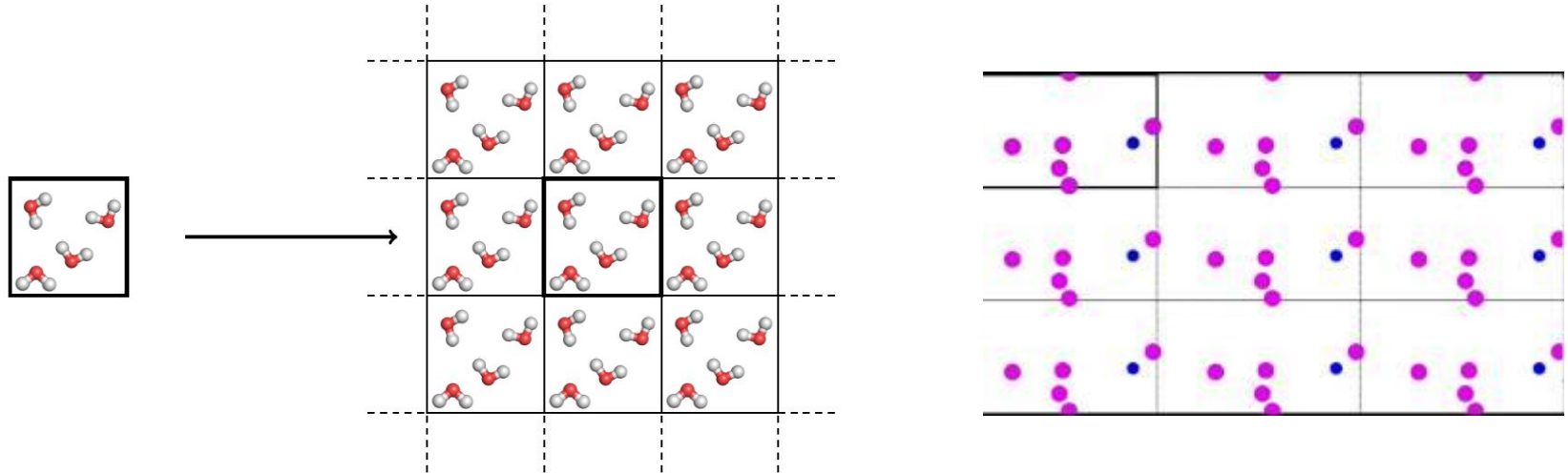
Observations?

- 2D
- LJ interactions
- Time steps
- Temperature
-

Missing from 3D

- Bonded and electrostatic interactions
- Fixed simulation box
- No PBC
-

Periodic Boundary Conditions



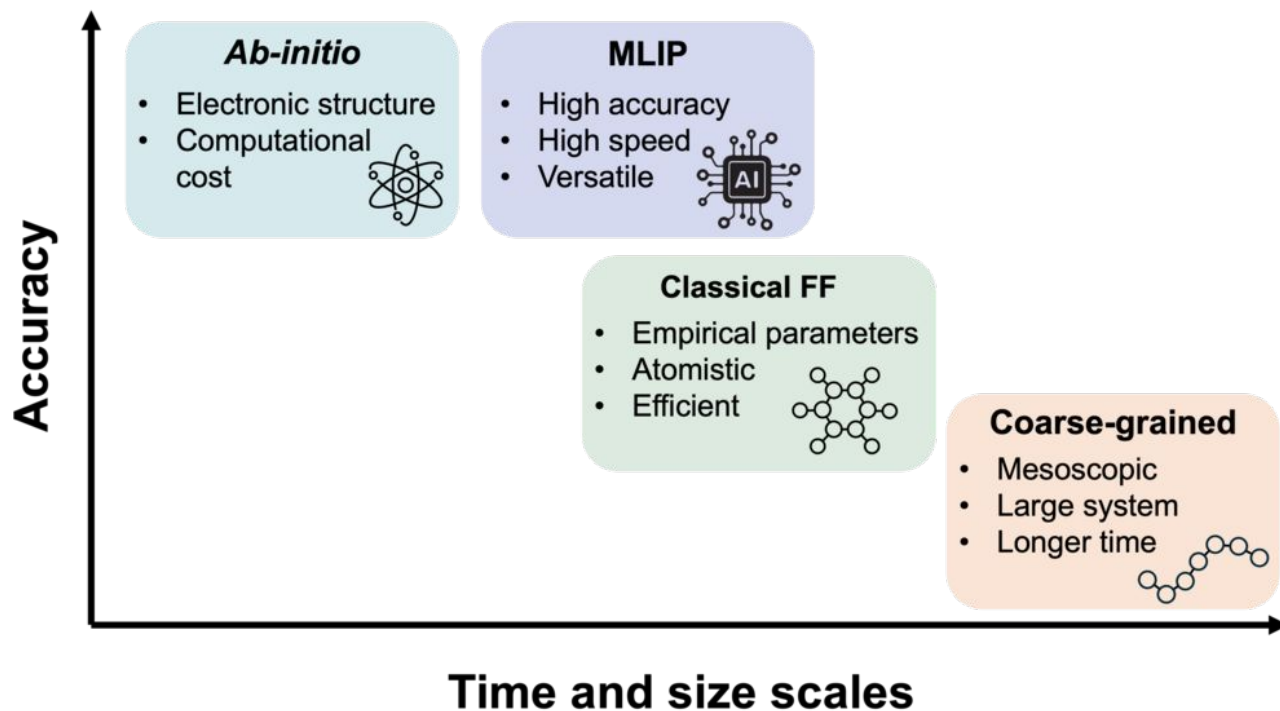
- Large systems are modeled using **Periodic Boundary Conditions (PBCs)**, creating an infinite array of unit cells.
- **Minimum-image convention:** each particle interacts only with the **nearest image** of all other particles.

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- Theory session (5:00-5:45 pm IST)
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 - **Force Field parameters**
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How to calculate forces (Interatomic potentials)

the accuracy and applicability of MD simulations is limited by the choice of **interatomic potential**, or also called "**force field**"



Choice of MD

Method	Description	Typical Scale	Application Examples
Ab-initio MD (AIMD)	Computes forces directly from quantum mechanics, enabling accurate treatment of bond formation, breaking, and electronic effects.	10^2 atoms, ps	Reaction mechanisms, proton transfer, bond breaking/forming
Classical Atomistic MD	Uses predefined force-field equations and parameters to model bonded and nonbonded interactions. Accuracy is highly dependent on the parameters	10^5–10^6 atoms, ns–μs	Polymers, biomolecules, liquids, materials stability
Coarse-Grained (CG) MD	Represents groups of atoms/molecules as single beads, drastically reducing the number of interaction sites. Enables the study of mesoscale behavior	10^6+ beads, μs–ms	Self-assembly, phase separation, micelles, polymer melts
ML Potentials (MLIPs)	Employ neural networks or other ML models trained on extensive DFT datasets to reproduce quantum-level interactions.	10^3–10^4 atoms, ns–μs	Defect dynamics, high-temperature behavior, reactive materials

Classical Force Field

The functional form of a typical forcefield

$$U_{\text{total}} = U_{\text{LJ}} + U_{\text{Coul}} + U_{\text{Bond}} + U_{\text{Angle}} + U_{\text{Dihedral}} + U_{\text{Improper}}$$

Non-bonded interactions

$$U_{\text{LJ}} = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right],$$

$$U_{\text{Coul}} = \frac{q_i q_j}{4\pi\epsilon_1\epsilon_0 r_{ij}},$$

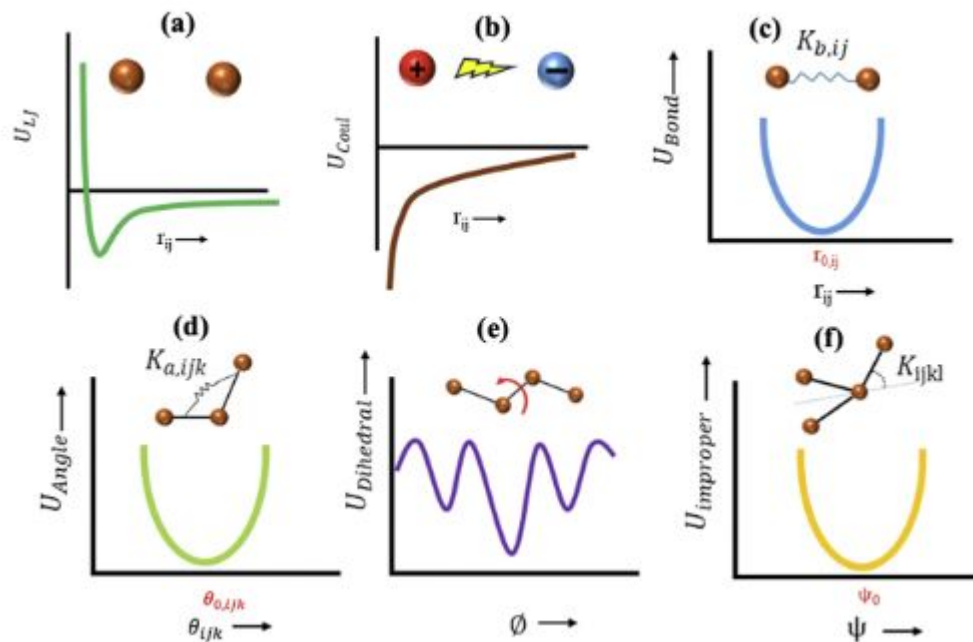
Bonded interactions

$$U_{\text{Bond}} = K_{b,ij}(r_{ij} - r_{0,ij})^2,$$

$$U_{\text{Angle}} = K_{a,ijk}(\theta_{ijk} - \theta_{0,ijk})^2,$$

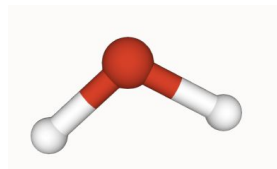
$$U_{\text{Dihedral}} = \frac{1}{2}K_1[1 + \cos(\phi)] + \frac{1}{2}K_2[1 - \cos(2\phi)] \\ + \frac{1}{2}K_3[1 + \cos(3\phi)] + \frac{1}{2}K_4[1 - \cos(4\phi)],$$

$$U_{\text{improper}} = K_{ijkl}(\psi - \psi_0)^2.$$



Classical Force Field

Guess the number of unique parameters for each molecule type

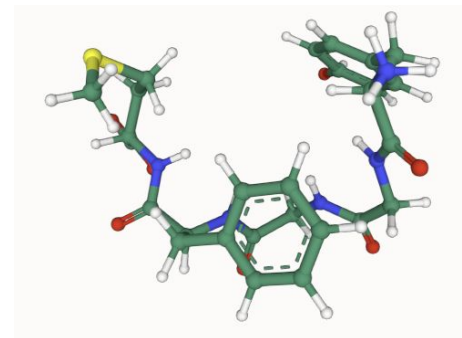
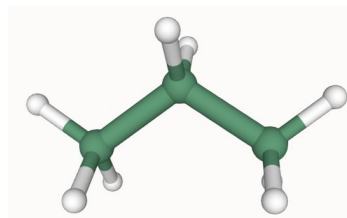
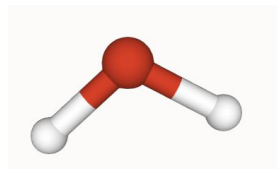


potential	parameter	value
van der Waals	σ	3.166 Å
van der Waals	ϵ	0.650 kJ/mol
bond	r_{OH}	1.000 Å
angle	$\angle_{\text{HOH}}$	109.47°
electrostatic	q_{O}	-0.8476e
electrostatic	q_{H}	$ q_{\text{O}} /2$

Unique parameters	Water	Propane	
Non-bonded	2		
Partial charges	2		
Bonds	1		
Angles	1		
Dihedrals/Impropers	0		
Total	6		

Classical Force Field

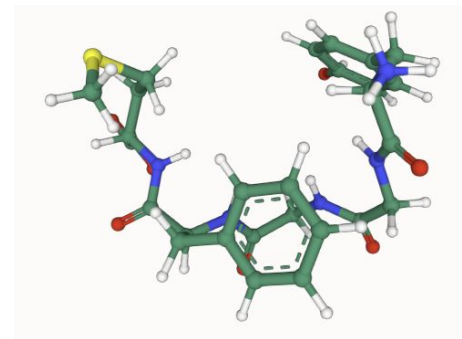
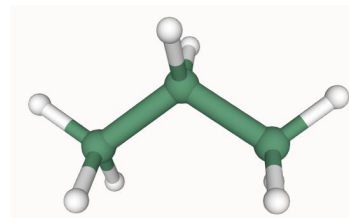
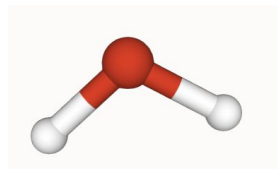
Guess the number of unique parameters for each molecule type



Unique parameters	Water	Propane	Small protein (Met-enkephalin)
Non-bonded	2	6	
Partial charges	2	3	
Bonds	1	4	
Angles	1	6	
Dihedrals/Impropers	0	12	
Total	6	31	

Classical Force Field

Guess the number of unique parameters for each molecule type



Unique parameters	Water	Propane	Small protein (Met-enkephalin)
Non-bonded	2	6	~30
Partial charges	2	3	~15
Bonds	1	4	~20
Angles	1	6	~50
Dihedrals/Impropers	0	12	~50
Total	6	31	~165

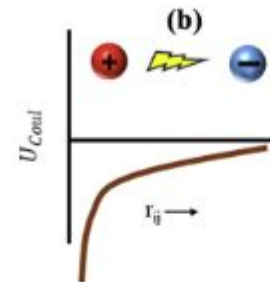
Common FFs

Force Field	Non-bonded	Bonded / Angle	Dihedral / Improper	Example Applications
OPLS-AA	LJ 12–6 + Coulomb	Harmonic	Fourier series for dihedrals; harmonic impropers	Organic liquids, polymers, small molecules
GAFF / AMBER	LJ 12–6 + Coulomb	Harmonic	Trigonometric cosine torsions; harmonic impropers	Small organic molecules, polymer precursors, biomolecules
CHARMM / CGenFF	LJ 12–6 + Coulomb; optional NBFIX	Harmonic	Multi-term torsions + Urey-Bradley; harmonic impropers	Biomolecules, organic molecules, polymers, membranes
COMPASS / PCFF	LJ 9–6 + Coulomb	Cubic–quartic	Valence–torsion cross terms; harmonic impropers	Polymers, condensed-phase materials, small molecules
GROMOS (UA)	LJ 12–6 + reaction-field Coulomb	Harmonic	Ryckaert–Belleman torsions; harmonic impropers	United-atom alkanes, biomolecules, polymer melts

OPLS and GAFF are the most broadly used for general organic polymers; CHARMM/CGenFF for bio-polymers; COMPASS/PCFF for engineering polymers; MARTINI for coarse-grained polymer systems.

Common FFs

Partial charges directly influences molecular geometry, stability, solvation, and the accuracy of simulated physical and thermodynamic properties.



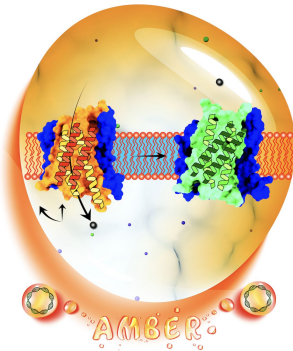
Force Field	Charge Assignment	Parameter Fitting / Target Properties
OPLS-AA	QM ESP fitting (HF/DFT); matches small-molecule dipoles	Optimized to reproduce condensed-phase properties such as liquid density, heat of vaporization, and dipole moments of small organic molecules.
GAFF / AMBER	RESP or AM1-BCC; compatible with biomolecular FF	Fitted to reproduce gas-phase quantum ESP , maintain compatibility with protein/nucleic acid force fields , and capture hydration/solvation energetics.
CHARMM / CGenFF	QM ESP + scaling; automated for new residues	Optimized to reproduce experimental condensed-phase data and maintain transferability across diverse chemical groups.
COMPASS / PCFF	Ab initio ESP of model compounds	Target properties include densities, cohesive energies, conformational energetics in solids and liquids.
GROMOS (UA)	Empirical or QM-derived; united-atom adjusted	Fitted to reproduce thermodynamic properties of liquids, and less focus on reproducing gas-phase ESP.

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Tools and Softwares

- **Python / Jupyter Notebook/Lab** – for workflow automation, running commands, visualising molecules/simulations, execute analysis scripts and plotting the data
- **AmberTools** – Parameterizing monomers using the Generalized Amber Force Field (GAFF) with AmberTools (including antechamber, prepgen, and tleap) to accurately define and construct polymer chains.
- **PACKMOL** – for generating initial multi-chain configurations
- **GROMACS** – performing standard MD equilibration and production simulations, including Energy Minimization (EM), NVT, and NPT simulations. Also, trajectory analysis.



FAST. FLEXIBLE. FREE.
GROMACS

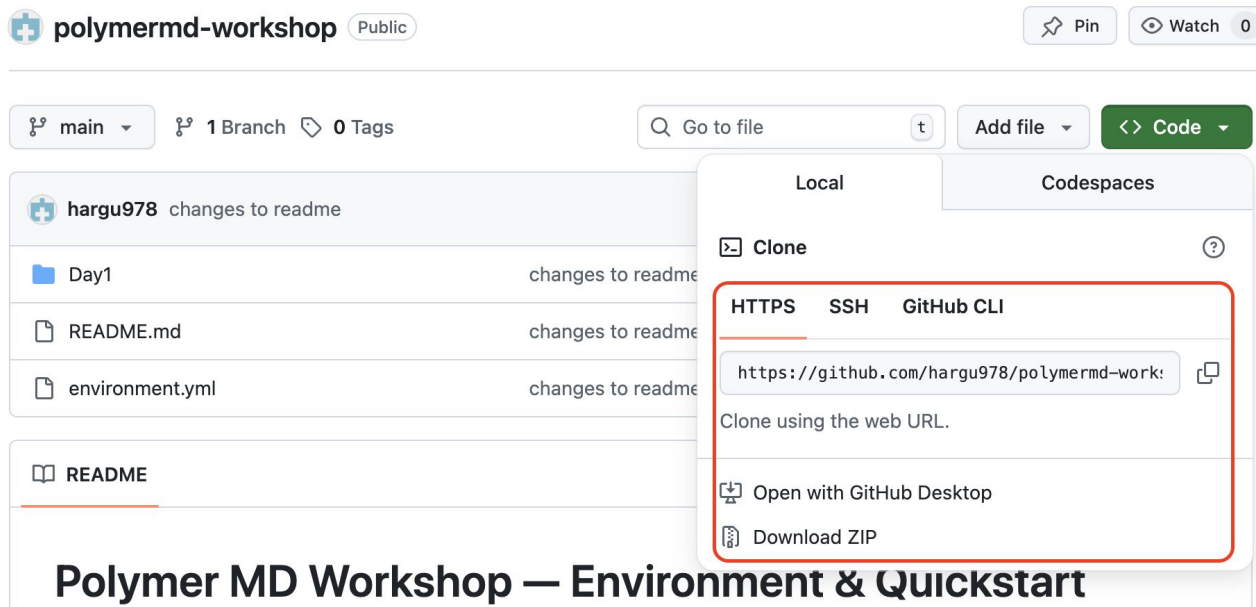
The GROMACS logo, featuring a stylized blue bird in flight.

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Installing the tools

- Get this repo: **hargu978/polymermd-workshop**
- If you have git installed:
git clone <https://github.com/hargu978/polymermd-workshop.git>
- Or download as ZIP file
- This contains only materials for today's workshop.
- Follow the steps on installing using **conda**.



polymermd-workshop Public

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hargu978 changes to readme

Day1 changes to readme

README.md changes to readme

environment.yml changes to readme

README

Polymer MD Workshop — Environment & Quickstart

Local Codespaces

Clone

HTTPS SSH GitHub CLI

<https://github.com/hargu978/polymermd-work:>

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

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ambertools

1. antechamber

- Generates atom types (GAFF/GAFF2)
- Computes partial charges (AM1-BCC, RESP)
- Produces parameter files (`.mol2`, `.prepi`, `.frcmod`)

2. parmchk2

- Checks for missing parameters
- Generates a `.frcmod` file containing supplemental parameters

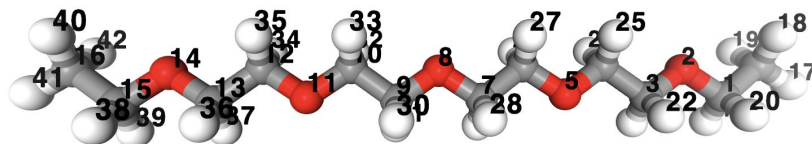
3. tleap

- Assembles the full system (molecule, solvent, ions)
- Writes AMBER `prmtop` and `inpcrd`

4. parmed

- Converts AMBER `prmtop` and `inpcrd` to GROMACS format

Monomer structure



Monomer GAFF atom typing and partial charge parameters

CHARGE	0.00 (0)								
Formula: H26 C12 O5									
ATOM	1	C	PEO	1	0.931	-0.097	-0.023	-0.097100	c3
ATOM	2	C1	PEO	1	2.445	-0.109	0.007	0.132400	c3
ATOM	3	O	PEO	1	2.928	0.653	-1.094	-0.428600	os
ATOM	4	C2	PEO	1	4.355	0.677	-1.114	0.127400	c3
ATOM	5	C3	PEO	1	4.829	1.500	-2.301	0.130400	c3
ATOM	6	O1	PEO	1	6.254	1.515	-2.312	-0.430600	os
ATOM	7	C4	PEO	1	6.750	2.278	-3.409	0.130400	c3
ATOM	8	C5	PEO	1	8.269	2.261	-3.377	0.129400	c3
ATOM	9	O2	PEO	1	8.762	3.024	-4.475	-0.429600	os
ATOM	10	C6	PEO	1	10.186	3.036	-4.489	0.129400	c3
ATOM	11	C7	PEO	1	10.660	3.858	-5.676	0.130400	c3
ATOM	12	O3	PEO	1	12.084	3.872	-5.692	-0.430600	os
ATOM	13	C8	PEO	1	12.575	4.634	-6.792	0.130400	c3
ATOM	14	C9	PEO	1	14.095	4.618	-6.764	0.127400	c3
ATOM	15	O4	PEO	1	14.598	5.379	-7.860	-0.428600	os

Day 2 - Outline

- Theory session (5:00-5:30 pm IST)
 - Recap
 - MD simulations ensembles
 - Polymer chain architecture
- Hands-on session (5:40-6:30 pm IST)
 - HEAD/CHAIN/TAIL atom definitions
 - Building polymer chains
 - Preparing input files